

Lenses

A lens is curved transparent glass that lets light through — bending it to magnify or shrink what we see.

Convex and concave lenses

A **lens** is a piece of transparent material (glass/plastic) with **curved surfaces**. Unlike mirrors, **light passes through** a lens — we see things *through* it.

Convex lens	Concave lens
Thicker in the middle , thin at edges	Thicker at the edges , thin in the middle
Converges light (converging lens)	Diverges light (diverging lens)
Close object → enlarged, erect (magnifier)	Always erect & diminished

Water-drop lens: a rounded drop of water over text acts like a convex lens and magnifies the letters — just like a magnifying glass.

Uses

- **Convex:** magnifying glass, spectacles for long sight, camera and the eye's own lens.
- **Concave:** spectacles for short sight, peepholes, some instruments.

Mirror vs lens: a mirror *reflects* light (we see *in* it); a lens *transmits* light (we see *through* it).

Where students lose marks

Trap 1 — Convex lens = **thick middle (converging)**; concave lens = **thick edges (diverging)**. Don't swap.

Trap 2 — We see **THROUGH** a lens, **not IN** it. Lenses transmit light; mirrors reflect it.

Trap 3 — A magnifying glass is a **convex lens** used with the object close to it.

Model answers — write them like this

Example A. Which lens is used as a magnifying glass and why?

Step 1: A magnifier needs an enlarged, erect image of a close object.

Step 2: A convex lens gives this when the object is close.

∴ A convex (converging) lens is used as a magnifying glass.

Example B. How can you tell a convex lens from a concave lens by touch/look?

Step 1: Feel the thickness: convex is thicker in the middle.

Step 2: Concave is thicker at the edges.

∴ Thick centre → convex; thick edges → concave.

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